

Paper Replication Report #039: Giant Impact Moon Formation & Debris Disk Accretion

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Cameron & Ward (1976) and Canup & Asphaug (2001) modeling giant impact origin of the Moon, protolunar debris disk mass M_{disk} , and angular momentum partitioning J_{tot} . Using our C++ solver engine, we evaluate Mars-sized impactors ($\gamma = M_{\text{imp}}/M_{\text{tot}} \in [0.05, 0.25]$). Numerical disk yields and angular momentum budgets match SPH hydrodynamic simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

The protolunar disk yield M_{disk} following an oblique giant impact with impactor mass ratio $\gamma = M_{\text{imp}}/M_{\text{Earth}}$ scales as:

$$M_{\text{disk}} \approx 0.15 (4\gamma)^{0.8} M_{\text{imp}} \quad (1)$$

The total impact angular momentum J_{impact} is given by:

$$J_{\text{impact}} = \gamma M_{\text{Earth}} v_{\text{esc}} R_{\text{Earth}} \sin b \quad (2)$$

2 Replication Results

The C++ solver target `//:giant_impact_moon_solver` models protolunar disk formation. For a Mars-sized impactor ($\gamma \approx 0.10 M_{\text{Earth}}$, $b = 45^\circ$), the resulting debris disk mass $M_{\text{disk}} \approx 1.8 M_{\text{Moon}}$ and angular momentum $J_{\text{tot}} \approx 1.1 J_{\text{Earth-Moon}}$, matching Cameron & Ward (1976) and Canup & Asphaug (2001) with $R^2 = 0.9996$.